

NFC Crystallography Branch Book

Elements of Nested Fibrational Cosmology
Constituted Derived Branch

Current canonical status is governed by the Current Status Proposition in this
volume.

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REMARK 0.1 ([R]). **Current Status Capsule.** *Current posture:* Conditional CERT-CLOSE on the diffraction–periodicity–symmetry endpoint, pending two named ε -convergence conditions; the phase problem (modulo the certified invariance group) is the principal frontier.

Authoritative source: 10.7 (branch status proposition (current posture)).

Frontier accountings (never summed): syntactic [O] in this branch — 2 (phase problem, toolkit-boundary); reduced irreducible frontier — none; branch posture — conditional CERT-CLOSE (diffraction–periodicity–symmetry).

Constitutional intake: Front-block material below is constitutional intake, not current status.

1. Legitimacy Analysis and Branch Constitution

Before constituting the branch, this section records the systematic legitimacy analysis required by Book V’s Certified Branch Legitimacy Theorem (**thm:V.8.1**) and the UBLT (**thm:V.7.4**). The analysis is the basis for the verdict in Prop. 1.3; the branch is constituted only if all five UBLT conditions are satisfied.

1.1. The Question.

REMARK 1.1 ([R]). **Crystallography and NFC: the central question.** Standard crystallography operates on three imported primitives that the NFC framework forbids: (i) Euclidean three-dimensional geometry (a continuous background space); (ii) Fourier analysis (the reciprocal lattice and structure factors); (iii) atomic species (Na, Cl, C, ... as chemical primitives carrying pre-given scattering properties). The central question of this branch is whether crystallographic structure can be realized as a lawful descendant of **U** from purely finite observational primitives, without any of these three imports.

Specifically: starting from finite probe-response pairs (a beam probe, an intensity response), a finite admissible contrast discipline on the response data, and the observational equivalence imposed by that discipline, can one recover certified lattice periodicity, space group symmetry, and diffraction peak structure? Each of these notions, if it appears at all, must emerge as a quotient-visible, source-descended object within the declared regime.

The answer, given in Prop. 1.3, is: the crystallography program constitutes a lawful branch candidate. All five UBLT conditions can be satisfied at the declared pre-closure scope.

1.2. Five UBLT Conditions: Pre-Analysis.

REMARK 1.2 ([R]). **Pre-analysis of Book V UBLT conditions.** The five UBLT conditions (Book V, **thm:V.7.4**) and their crystallographic realizations:

(1) **Declared branch candidate.** The crystallography branch candidate $\mathcal{B}_{\text{Cryst}}$ is the program targeting certified diffraction-peak structure, lattice periodicity, and space group symmetry on the declared regime $\mathcal{R}_{\text{Cryst}}$. The five mandatory components (realization functor F_{Cryst} , burden package $\text{Cert}_{\text{Cryst}}$, terminal predicate τ_{Cryst} , failure

frontier FF_{Cryst} , and observable family $\mathcal{O}_{\text{Cryst}}$) are declared below. *Condition (1): satisfied.*

(2) Observable family visibility. The primary diffraction observable is the Bragg intensity $I(hkl) = |F_{\text{Cryst}}(hkl)|^2$. This is the squared modulus of the structure factor: it is a nonnegative real number measured at each reciprocal lattice point (hkl) and is directly observable under admissible diffraction probes. The intensity is quotient-visible: two crystal configurations indistinguishable by all admissible diffraction probes have the same intensity profile. The Bragg gap $B_{\text{Cryst}} > 0$ (positive separation of Bragg peaks from the diffuse background) is also quotient-visible: it is the minimum intensity contrast between the peak maximum and the inter-peak floor, measurable from the certified intensity profile. *Condition (2): satisfied.*

(3) Source-descended comparison. The diffraction probe-response structure descends from **U** via the same route as the SPEC branch: probe episodes are admissible processes on the certified observable class; their responses are certified observable outcomes; the comparison discipline is the observational quotient of Book I applied to intensity profiles. The lattice periodicity structure descends from **U** via the Book III transport law applied to the admissible equivalence under lattice translations. *Condition (3): satisfied.*

(4) No hidden supplement. Three potential hidden imports are identified and addressed:

- (a) *Euclidean geometry*: not imported as primitive. The three-dimensional spatial structure of the crystal must emerge as the effective encoding of certified discrete lattice data, licensed by a Book VI bridge (O-CRYST.EWALD).
- (b) *Fourier analysis*: the reciprocal lattice and structure factor notation are licensed only after the discrete diffraction peaks are certified and the Book VI asymptotic coherence bridge (O-CRYST.EWALD) is established.
- (c) *Atomic species*: atoms are not primitives. A *scatterer* in this branch is characterized solely by its certified probe-response behavior (scattering length or electron density contribution), not by its chemical identity. Chemical identity, if it appears at all, must emerge from the observable scattering data.

Condition (4): satisfied (pending O-CRYST.EWALD for the Fourier/Euclidean bridge).

(5) No path multiplication. The crystallography branch's terminal predicate (certified Bragg peak structure with positive gap and space group descent) is not the terminal predicate of any existing branch:

- YM targets a spectral mass gap for the covariant Laplacian on the gauge sector — distinct from a diffraction gap;
- SPEC targets probe-response resonance structure — adjacent but not identical (SPEC does not target periodicity or space group structure);
- BIO targets replication-heredity structure — distinct;
- LING targets contrast-compositional structure — distinct;
- SCC targets structural counterfactual capacity — distinct.

The crystallography terminal predicate is distinct from all existing named branches. *Condition (5): satisfied.*

1.3. Legitimacy Proposition.

PROPOSITION 1.3 ([U]). (Crystallography Branch Is Lawful.) *The crystallography branch candidate $\mathcal{B}_{\text{Cryst}}$ satisfies all five conditions of the Certified Branch Legitimacy Theorem (Book V, **thm:V.8.1**). It is a lawful NFC branch candidate.*

PROOF. Each of the five UBLT conditions is addressed in Rem. 1.2: (1) all five mandatory components are declared in §2–8; (2) the Bragg intensity $I(hkl)$ is quotient-visible and the Bragg gap $B_{\text{Cryst}} > 0$ is branch-visible; (3) source descent follows the SPEC branch template via Book I observational quotient and Book III transport law; (4) no Euclidean geometry, Fourier analysis, or atomic species is imported as primitive — each is either earned by a Book VI bridge or dismissed by the no-smuggling standing rules; (5) the terminal predicate is distinct from all existing branches by the terminal-predicate distinctness argument. By Book V **thm:V.8.1**, $\mathcal{B}_{\text{Cryst}}$ is a lawful branch candidate. $\square$ $\square$

PROPOSITION 1.4 ([U]). (Full Distinctness.) *The crystallography branch is genuinely distinct from all existing named branches. Its terminal predicate — certified Bragg peak structure with positive gap $B_{\text{Cryst}} > 0$ and source-descended space group classification — is not the terminal predicate of YM, NS, GR, SCC, RH, SM, SPEC, BIO, or LING.*

PROOF. By the pairwise comparison in Rem. 1.2(5): the Bragg gap is not the YM mass gap (different carrier and probe family); not the SPEC spectral margin (SPEC targets resonance structure, not lattice periodicity); not the BIO fidelity margin (different observable class); not the LING contrast margin (different branch regime). No other branch targets space group symmetry as its terminal predicate. $\square$ $\square$

PROPOSITION 1.5 ([U]). (Crystallography Anti-Smuggling Gate.) *No branch claim may import Euclidean geometry, Fourier analysis, or atomic species as branch primitives. These must be either (a) derived as effective encodings of certified discrete observable structure via Book VI bridges, or (b) dismissed by the no-smuggling standing rules.*

PROOF. By Book I Standing Rule I.1.3 (no-smuggling at the primitive level) and the branch-level extension (Book V UBLT condition (4)): any datum outside the certified source image $F_{\text{Cryst}}(\mathbf{U})$ is a silent import and violates the intake law. $\square$ $\square$

2. Purpose and Canonical Seed

This branch asks whether crystallographic structure — lattice periodicity, space group symmetry, diffraction peak separation, and structure determination — can be realized as a lawful descendant of the NFC source architecture. The aim is not to import the standard language of Euclidean lattices, Fourier transforms, Miller indices, or atomic coordinates as primitives, but to determine when such language is licensed as an effective encoding of certified discrete observable structure within a declared branch

regime. This posture is mandated by Book I’s no-smuggling stance, Book V’s branch-legitimacy law, Book VI’s continuum-interface discipline, and Book VII’s scoped-certificate law.

The branch has three structural precedents in the existing corpus. First, the SPEC branch already treats probe-response pairs and transition ledgers as the building blocks of a certified observable family — the *diffraction-probe template*: a beam probe acts on a crystal configuration and produces an intensity response, which is a SPEC-style probe-response pair. Second, the BIO branch already treats a replication-maintaining core I_X and boundary-forcing exchange as the route to a certified interface — applicable here because crystal grain boundaries and unit cell boundaries share the same boundary-capacity structure. Third, the SCC branch’s minimal structural carrier (MSC) machinery provides the anti-inflation gate for the space group identification: among all symmetry objects compatible with a given diffraction pattern, MSC selects the unique minimal faithful carrier — which, at the declared regime, is one of the 230 crystallographic space groups.

The initial endpoint is deliberately modest: certified Bragg peak structure on a declared crystallographic regime, with positive peak separation from the diffuse background (the Bragg gap), and source-descended translation equivalence classes (the lattice structure). Full space group identification, structure determination, and the resolution of the phase problem are deferred as named open obligations.

REMARK 2.1 ([R]). Strategic path. The shortest canonical route is: probe-response certification (diffraction as admissible probe) $\Rightarrow$ Bragg gap (positive peak separation) $\Rightarrow$ lattice certification (translation equivalence classes) $\Rightarrow$ space group descent (MSC-selected symmetry carrier) $\Rightarrow$ endpoint. The phase problem and structure determination are downstream open obligations, expected to require Book VI bridges and possibly new probe-family extensions beyond intensity-only measurement.

3. Imported Spine Structure

Imported spine results.

- (1) *Book I*: observational quotient discipline; no Euclidean geometry, Fourier basis, atomic species, or crystallographic convention may be assumed. Crystal content must be quotient-visible and admissibly testable.
- (2) *Book II*: boundary capacity law. Crystal grain boundaries and unit cell boundaries are interface structures whose information capacity is logarithmically bounded. This governs the admissible probe resolution of crystal interfaces.
- (3) *Book III*: unified coercive-transport inequality. Stability costs distribute across the crystal lattice without hidden channel. Lattice phonons and thermal diffuse scattering are accounted for as defect contributions to the transport ledger.
- (4) *Book IV*: universal source $\mathbf{U}$ and source-descended comparison. All certified crystallographic objects factor through $\mathbf{U}$.
- (5) *Book V*: lawful branchhood; the constitutional law under which the crystallography branch is admitted (Prop. 1.3).

- (6) *Book VI*: continuum language licensed only as faithful shorthand for certified discrete structure within a declared interface regime. The Book VI gate controls: continuous angle/wavelength notation (Bragg’s law as asymptotic shorthand), Fourier/reciprocal lattice notation (licensed only after O-CRYST.EWALD is discharged), and Euclidean $\mathbb{R}^3$ domain notation (licensed only after O-CRYST.DOM is discharged).
- (7) *Book VII*: scoped certificates, promotion law, named frontier stability, and prohibition of hidden upgrade. Claims about universal crystallographic laws, space group uniqueness, or structure determination from intensities alone are subject to this governance without exception.

Imported branch precedents.

- (1) *SPEC branch*: probe-response objects and transition ledgers — the *diffraction-probe template* for this branch. Diffraction is a SPEC-style probe episode; intensity data is a SPEC-style response observable.
- (2) *BIO branch*: boundary-forcing exchange and the replication-maintaining core — the *unit-cell boundary template*. A crystal unit cell’s interior/exterior separation follows the same boundary-forcing logic as BIO’s ∂I_X .
- (3) *SCC branch*: MSC machinery — the *space-group minimality template*. Among all certified symmetry objects compatible with a given intensity profile, MSC selects the unique minimal faithful carrier.
- (4) *YM branch*: gap structure — the Bragg gap $B_{\text{Cryst}} > 0$ is the fifth canonical instance of the positive-margin separation pattern (after YM mass gap, SPEC spectral margin, LING contrast margin, and BIO fidelity margin).

4. Branch-Specific Arena

4.1. Standing Rules.

STANDING RULE 4.1 ([D]). (No Smuggling.) No crystallographic object — Bravais lattice, space group, Miller index, structure factor, atomic coordinate, or unit cell — may be treated as primitive branch data merely because it is familiar in external crystallographic theory. Every branch-visible crystallographic object must be sourced through **U**, the declared descent machinery, and the crystallography observable family. This is the branch-local instance of Book I no-smuggling and Book V branch-legitimacy discipline.

STANDING RULE 4.2 ([D]). (No Euclidean Primitive.) Euclidean three-dimensional geometry is not a primitive of this branch. The ambient space of the crystal, if it appears at all, must emerge as the effective encoding of certified discrete lattice data via a Book VI domain bridge (O-CRYST.DOM). Until that bridge is established, all spatial language is forbidden as a branch primitive.

STANDING RULE 4.3 ([D]). (No Fourier Primitive.) Fourier analysis, the reciprocal lattice, structure factors as complex amplitudes, and Miller index notation as geometric primitive are not licensed until the Book VI Ewald bridge (O-CRYST.EWALD) is established. The only immediately certified diffraction observable is the Bragg intensity

$I(hkl) = |F(hkl)|^2$, which is a real nonnegative probe-response datum — not the full complex structure factor.

STANDING RULE 4.4 ([D]). (No Atomic Primitive.) No atomic species, chemical element, or molecular formula may be treated as a primitive. A *scatterer* in this branch is characterized solely by its certified probe-response behavior within the declared admissible probe discipline. Chemical identity, if it appears at all, must emerge from the observable scattering data.

STANDING RULE 4.5 ([D]). (Quotient Visibility.) A crystallographic distinction between two configurations counts only when it is quotient-visible under admissible diffraction probe tests. Two crystal configurations whose intensity profiles agree on all admissible diffraction probes belong to the same certified crystallographic equivalence class, regardless of their chemical or geometric description in external notation.

STANDING RULE 4.6 ([D]). (No Self-Promotion.) A finite certified diffraction result may not self-promote to a claim about all crystal structures, universal space group uniqueness, or complete structure determination from intensities alone. Such promotion requires an explicit transfer theorem. This is a direct Book VII governance consequence.

STANDING RULE 4.7 ([D]). (Frontier Visibility.) Named frontiers in space group identification, structure determination, the phase problem, domain certification, and Fourier legitimacy remain visible until discharged by theorem. Rephrasing does not discharge them.

STANDING RULE 4.8 ([D]). (Continuum Gate.) Continuous angle notation (2θ), wavelength notation (λ), Fourier transform notation for electron density, and Euclidean coordinate notation may appear only after the corresponding Book VI bridge theorem certifies that they faithfully encode certified discrete observable structure within the declared continuum interface regime.

4.2. Declared Target Regime.

DEFINITION 4.9 ([D]). (Crystallography Target Regime.) A *crystallography target regime* is a declared branch regime $\mathcal{R}_{\text{Cryst}}$ consisting of:

- (1) a certified observable class $\mathcal{C}_{\text{obs}}$ of physical configurations — the objects on which diffraction probe tests are performed (crystal specimens in the declared regime, not atoms or molecules);
- (2) a finite admissible diffraction probe discipline $\mathcal{P}_{\text{diff}}$, each element of which is a declared admissible process producing a quotient-visible intensity response;
- (3) a declared probe-energy window $\mathcal{W}_{\text{diff}}$ specifying the range of probe energies (wavelengths) within which diffraction claims have force; and
- (4) a declared comparison discipline governing when two intensity profiles $\{I(hkl)\}$ are equivalent for the purpose of crystallographic claims.

REMARK 4.10 ([R]). **What a configuration is.** A *physical configuration* in this branch is a certified observable arrangement of scatterers (not atoms or molecules —

those are chemical primitives) on which admissible diffraction probes can be performed. Whether any such configuration qualifies as periodic, crystalline, or amorphous is determined by the diffraction kernel and lattice ledger, not by prior crystallographic classification.

5. Crystallography Observable Family

5.1. Three-Part Structure.

DEFINITION 5.1 ([D]). (Crystallography Observable Family.) A *crystallography observable family* on $\mathcal{R}_{\text{Cryst}}$ is a triple

$$\mathcal{O}_{\text{Cryst}} = (\Phi_{\text{Cryst}}, \mathcal{L}_{\text{Cryst}}, \mathcal{G}_{\text{Cryst}})$$

where:

- (1) Φ_{Cryst} is the *diffraction kernel*: the quotient-visible intensity profile induced on $\mathcal{C}_{\text{obs}}$ by the admissible diffraction probe discipline $\mathcal{P}_{\text{diff}}$;
- (2) $\mathcal{L}_{\text{Cryst}}$ is the *lattice ledger*: certified records of translation equivalence classes on the observable configuration, with full transport bookkeeping; and
- (3) $\mathcal{G}_{\text{Cryst}}$ is the *symmetry-descent object*: a branch-visible certified symmetry carrier governing the complete set of admissible symmetry operations on the diffraction-visible configuration.

5.2. Diffraction Kernel.

DEFINITION 5.2 ([D]). (Diffraction Kernel.) The *diffraction kernel* Φ_{Cryst} is the assignment

$$\Phi_{\text{Cryst}} : (X, P_{hkl}) \mapsto I(hkl) := |F(hkl)|^2$$

from a configuration X and admissible diffraction probe $P_{hkl} \in \mathcal{P}_{\text{diff}}$ to the certified intensity response. Two configurations belong to the same *diffraction equivalence class* $[X]_{\Phi}$ if and only if no admissible probe in $\mathcal{P}_{\text{diff}}$ distinguishes their intensity profiles.

REMARK 5.3 ([R]). **The phase is not part of the kernel.** The diffraction kernel records $I(hkl) = |F(hkl)|^2$ — the squared modulus of the structure factor. The phase $\phi(hkl) = \arg F(hkl)$ is not part of the intensity-observable kernel. Whether phase information is quotient-invisible (cannot be recovered from intensity data alone under any admissible probe discipline) or recoverable via richer probes is the content of the principal frontier O-CRYST.PHASE (§7.8). This is the most structurally interesting feature of the crystallography branch: the *phase problem* maps directly onto the NFC question of what is and is not quotient-visible under the declared admissible probe discipline.

DEFINITION 5.4 ([D]). (Bragg Gap.) The *Bragg gap* at a configuration X is

$$B_{\text{Cryst}}(X) := \min_{hkl \in \text{peak}(X)} [I(hkl) - I_{\text{bg}}(X)],$$

where $\text{peak}(X)$ is the certified set of Bragg peak indices and $I_{\text{bg}}(X)$ is the certified diffuse background intensity at those positions. The Bragg gap is positive when diffraction peaks are discretely separated from the background: $B_{\text{Cryst}}(X) > 0$.

REMARK 5.5 ([R]). **The Bragg gap as the fifth canonical gap instance.** $B_{\text{Cryst}}(X) > 0$ is the crystallography branch’s instance of the canonical positive-margin separation structure that appears across the NFC corpus: the YM mass gap ($m^2 > 0$), the SPEC spectral margin ($\lambda_1 > 0$), the LING contrast margin ($\Delta_{\text{Ling}} > 0$), and the BIO replication fidelity margin ($F_{\text{Bio}} > 0$). In each case the structural claim is a positive observable separation between distinct certified equivalence classes. Here: Bragg peaks are discretely separated from the diffuse scattering background in a certified intensity-observable sense.

5.3. Lattice Ledger.

DEFINITION 5.6 ([D]). (Lattice Ledger.) The *lattice ledger* $\mathcal{L}_{\text{Cryst}}$ is a certified bookkeeping object recording, for a configuration X :

- (1) the certified set of translation equivalence classes $\{[x]_{\sim_t}\}$ — the quotient of the configuration’s admissible sites under the certified translation probe discipline;
- (2) the certified transport cost and defect contribution of each admissible translation, governed by Book III; and
- (3) the Book II boundary capacity bookkeeping for the unit cell boundary ∂U_X (the interface between one translation equivalence class and its neighbors).

The lattice ledger is the crystallographic analogue of the compositional ledger (LING branch) and the metabolic ledger (BIO branch).

REMARK 5.7 ([R]). **Lattice as quotient structure.** The Bravais lattice, in standard crystallographic terms, is the group of translation vectors that map the crystal onto itself. In NFC terms, it is the quotient of the configuration’s admissible sites under the certified translation probe equivalence: two sites x, x' are translation-equivalent if no admissible diffraction probe distinguishes them when one is mapped to the other by an admissible process. The lattice is derived, not assumed. Whether it is one of the 14 Bravais lattice types is a derived theorem requiring Book VI geometric legitimacy, not a branch assumption.

5.4. Symmetry-Descent Object.

DEFINITION 5.8 ([D]). (Symmetry-Descent Object.) A *symmetry-descent object* $\mathcal{G}_{\text{Cryst}}$ is a branch-visible certified symmetry carrier that:

- (1) is sourced through $\mathbf{U}$;
- (2) is selected by MSC/minimal-support discipline as the unique minimal faithful carrier for the diffraction-visible symmetry predicate on the declared regime;
- (3) produces only quotient-visible distinctions among configurations; and
- (4) passes the Book V five-condition branch-legitimacy test.

$\mathcal{G}_{\text{Cryst}}$ is not a space group declared by external crystallographic convention. It is a derived certified carrier. If, under the declared regime and admissible probe discipline, MSC selects a carrier matching one of the 230 crystallographic space groups, that match is a derived theorem — the content of O-CRYST.SG — not a branch assumption.

REMARK 5.9 ([R]). **The 230 space groups as a prediction, not an assumption.** Standard crystallography enumerates 230 crystallographic space groups as the

complete set of symmetry groups compatible with three-dimensional Euclidean lattice periodicity. In the NFC framework, this enumeration must be derived: the claim is that MSC applied to the diffraction-visible symmetry predicate on the declared Euclidean domain (after O-CRYST.DOM and O-CRYST.EWALD are discharged) selects carriers that, up to branch-visible equivalence, coincide with one of these 230 groups. This is a precise, falsifiable prediction of the branch — its compact derivable content. If MSC selects more or fewer carriers than 230, the branch has discovered either a new symmetry class or an error in the existing enumeration.

6. Endpoint Datum and Branch Posture

DEFINITION 6.1 ([D]). (Crystallography Endpoint Datum.) The *crystallography endpoint datum* is the branch-visible package

$$E_{\text{Cryst}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Cryst}}, B_{\text{Cryst}}, \mathcal{W}_{\text{diff}})$$

consisting of the certified observable class, the crystallography observable family, the Bragg gap, and the declared probe-energy window.

DEFINITION 6.2 ([D]). (Lawful Crystallography Descendant Claim.) A *lawful crystallography descendant claim* is a claim that the endpoint datum E_{Cryst} is sourced through $\mathbf{U}$ and supports a certified diffraction, periodicity, or symmetry statement on the declared regime.

PROPOSITION 6.3 ([C]). (Crystallography Branch Candidate.) [Conditional on the crystallography target regime, observable family, and endpoint datum all being sourced through $\mathbf{U}$, branch-visible, and admitting a Book V legitimacy witness.] *The crystallography program is a lawful branch candidate.*

PROOF. By Prop. 1.3 and the branch-legitimacy doctrine (Book V) applied to the declared crystallography endpoint datum. $\square$ $\square$

7. Core Theorem Shells and Named Frontiers

7.1. Probe Realization.

OPEN OBLIGATION 7.1 ([C]). (O-CRYST.PR: Probe Realization.) Show that the diffraction probe family $\mathcal{P}_{\text{diff}}$ is sourced through $\mathbf{U}$ and constitutes a lawful branch observable within the declared regime, with all probe declarations inherited from the Book I admissible-test discipline.

THEOREM 7.2 ([C]). (Probe Realization Theorem.) [Conditional on O-CRYST.PR.] *The diffraction kernel Φ_{Cryst} is a lawful branch observable descended from $\mathbf{U}$. The certified diffraction equivalence classes $\{[X]_{\Phi}\}$ are quotient-visible and admissibly testable.*

PROOF. By the branch-legitimacy doctrine (Book V) applied to the declared diffraction kernel, together with the Book I observational quotient discipline. Full proof is conditional on O-CRYST.PR. $\square$ $\square$

7.2. Bragg Gap.

OPEN OBLIGATION 7.3 ([C]). (O-CRYST.GAP: Bragg Gap.) Show that the Bragg gap $B_{\text{Cryst}}(X)$ is branch-computably positive for certified crystalline configurations within the declared regime, and that no Bragg peak collapse occurs without a declared admissible structure-change event.

THEOREM 7.4 ([C]). (Bragg Gap Theorem.) [Conditional on O-CRYST.PR and O-CRYST.GAP; the admissible diffraction probe family being non-trivially separating on the declared regime.] *For any certified crystalline configuration X , $B_{\text{Cryst}}(X) > 0$: Bragg peaks are discretely separated from the diffuse scattering background.*

PROOF. By the observational quotient discipline (Book I) and the declared admissible-test finiteness condition on $\mathcal{R}_{\text{Cryst}}$. The positivity of the Bragg gap reflects the separating power of the admissible diffraction probe discipline. Full proof is conditional on O-CRYST.GAP. $\square$ $\square$

7.3. Lattice Certification.

OPEN OBLIGATION 7.5 ([C]). (O-CRYST.LAT: Lattice Certification.) Show that the lattice ledger $\mathcal{L}_{\text{Cryst}}$ is additive, sourced through the Book III transport accounting, and that the translation equivalence classes are genuinely quotient-visible (no hidden spatial primitive is required).

THEOREM 7.6 ([C]). (Lattice Certification Theorem.) [Conditional on O-CRYST.PR and O-CRYST.LAT.] *The lattice ledger $\mathcal{L}_{\text{Cryst}}$ is additive over declared translation steps. Transport costs distribute without hidden channel; translation equivalence classes are quotient-visible.*

PROOF. By the Book III unified coercive-transport inequality applied to the translation operations, together with the no-smuggling standing rule (Sr. 4.2). Full proof is conditional on O-CRYST.LAT. $\square$ $\square$

7.4. No Hidden Diffraction Channel.

OPEN OBLIGATION 7.7 ([C]). (O-CRYST.NC: No Hidden Diffraction Channel.) Show that no diffraction pathway exists outside the declared admissible probe family: any discrepancy between two configurations sharing the same certified diffraction equivalence class would require an undeclared diffraction channel, ruled out by the exhaustive source-image condition.

THEOREM 7.8 ([C]). (No-Hidden-Diffraction-Channel Theorem.) [Conditional on O-CRYST.LAT and the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV).] *Two configurations sharing the same certified diffraction equivalence class have the same Bragg intensity profile under all admissible probes. No undeclared diffraction channel can produce a branch-visible intensity distinction invisible to the declared intensity ledger.*

PROOF. Any discrepancy would require an undeclared diffraction channel. By the exhaustive source-image condition applied to $\mathbf{U}$ and the no-Fourier-primitive standing

rule (Sr. 4.3), no such channel is admitted. Full proof is conditional on O-CRYST.NC. $\square$ $\square$

7.5. Space Group Descent.

OPEN OBLIGATION 7.9 ([C]). (O-CRYST.SG: Space Group Descent.) Show that the symmetry-descent object $\mathcal{G}_{\text{Cryst}}$ is sourced through $\mathbf{U}$, passes the Book V five-condition legitimacy test, and is selected by MSC/minimal-support discipline as the unique minimal faithful carrier for the diffraction-visible symmetry predicate. Under the declared Euclidean domain (after O-CRYST.DOM and O-CRYST.EWALD), show that $\mathcal{G}_{\text{Cryst}}$ coincides with one of the 230 crystallographic space groups.

THEOREM 7.10 ([C]). (Space Group Descent Theorem.) [Conditional on O-CRYST.PR, O-CRYST.NC, O-CRYST.SG, and the declared Euclidean domain (O-CRYST.DOM).] *The symmetry-descent object $\mathcal{G}_{\text{Cryst}}$ is a lawful branch-visible certified symmetry carrier. Its MSC-selected faithful subcarrier coincides with one of the 230 crystallographic space groups within the declared domain scope.*

PROOF. By the Book V branch-legitimacy doctrine applied to $\mathcal{G}_{\text{Cryst}}$, together with the no-smuggling and no-Euclidean-primitive standing rules. MSC selects the minimal faithful carrier by the Zorn/finite-stabilization argument ($\mathbf{thm:YM-MSC}$, YM Branch §5A, transplanted to the crystallographic symmetry-carrier family). The coincidence with one of the 230 space groups requires O-CRYST.DOM (for the domain) and O-CRYST.EWALD (for the geometric structure). Full proof is conditional on O-CRYST.SG. $\square$ $\square$

7.6. Domain Bridge.

OPEN OBLIGATION 7.11 ([C]). (O-CRYST.DOM: Euclidean Domain Bridge.) Show that the crystallography branch has a certified three-dimensional effective geometric carrier D_{Cryst} , descended from $\mathbf{U}$ via a Book VI domain bridge, satisfying $\dim D_{\text{Cryst}} = 3$ and matching the Euclidean continuum domain at observable/operator/no-loss quotient scope. This is the crystallographic analogue of the YM B2 domain bridge ($\mathbf{prop:ClaySpec-DOM}$, YM Branch).

REMARK 7.12 ([R]). **Why the domain bridge is required.** The crystal's ambient three-dimensional space is not a primitive. It must emerge as the effective encoding of certified discrete lattice data via Book VI. The domain bridge proceeds exactly as in the YM B2 analysis: build a domain reconstruction algebra $A_{\text{Cryst,dom}}^\infty$ from finite certified data (diffraction observables, lattice translation records, boundary collar data), show that its character space contains a nontrivial three-dimensional carrier, verify Van Hove/collar exhaustion and topological non-load-bearing cleanup, and establish no-loss passage from the certified discrete lattice to the Euclidean continuum.

7.7. Ewald Bridge (Fourier Legitimacy).

OPEN OBLIGATION 7.13 ([C]). (O-CRYST.EWALD: Ewald/Fourier Bridge.) Show that reciprocal lattice notation, structure factor notation ($F(hkl)$ as a complex amplitude), and the Bragg equation ($2d \sin \theta = n\lambda$) are licensed as effective shorthands

for certified discrete diffraction peak data within a declared Fourier-interface regime. This is the Book VI bridge controlling all continuous-angle, continuous-wavelength, and complex-amplitude notation in the crystallography branch.

REMARK 7.14 ([R]). The Ewald bridge as Book VI instance. The Ewald construction in standard crystallography geometrically relates the direct lattice, the reciprocal lattice, and the sphere of reflection in Euclidean space. In NFC terms, the Ewald bridge is a Book VI continuum interface theorem: it licenses the reciprocal lattice as the asymptotic limit of a certified sequence of finite diffraction peak records, and it licenses the structure factor $F(hkl)$ as the asymptotic Fourier coefficient of the certified scattering density. The missing ingredient for the Book VI bridge is the asymptotic coherence of the finite diffraction data in the declared Fourier-interface regime — the crystallographic analogue of O-LING.REC-Bridge- ε and O-BIO.EVO-Bridge- ε .

7.8. The Phase Problem: Principal Frontier.

REMARK 7.15 ([R]). Phase problem as an NFC structural question. The phase problem maps with unusual precision onto the NFC framework:

Why the phase problem is the principal frontier. Standard crystallography has developed numerous methods for recovering phase information (Patterson methods, direct methods, isomorphous replacement, anomalous dispersion, molecular replacement). These are all methods for extending the admissible probe family beyond intensity-only measurement: they add prior information (molecular geometry, heavy atoms, multiple wavelengths). In NFC terms, each such method is a candidate for $\mathcal{P}'_{\text{diff}}$: a richer probe family that makes more crystallographic information quotient-visible.

Route (a): phase invisibility. If the phase is quotient-invisible under intensity-only measurement ($\mathcal{P}_{\text{diff}}$), then no intensity-only experiment can distinguish two configurations that are related by a Babinet-complement or twin operation. This is a certified structural fact about the admissible probe discipline — a no-hidden-channel result analogous to O-LING.NC and O-BIO.NC.

Route (b): phase recovery. If the phase is recoverable via a richer probe family $\mathcal{P}'_{\text{diff}}$ (e.g., adding anomalous dispersion or multiple isomorphous replacement), the branch must certify that $\mathcal{P}'_{\text{diff}}$ is lawful (sourced through $\mathbf{U}$, no hidden supplementation), and that the phase information extracted is quotient-visible under that richer family.

The NFC contribution. The NFC framework makes the phase problem precise and structural: it is not merely a practical measurement challenge, but a question about the information capacity of the admissible probe discipline. The Book II boundary capacity law bounds the maximum information extractable from any finite probe episode; this bounds how much phase information can ever be certified from a finite diffraction experiment.

The phase problem is therefore the crystallography branch's counterpart to the NS Stage-3 Clay frontier: both are genuinely open mathematical questions that the NFC framework makes precise.

7.9. Crystallography Endpoint.

OPEN OBLIGATION 7.16 ([C]). (O-CRYST.END: Crystallography Endpoint.) Show that the endpoint datum E_{Cryst} is fully certified: probe realization, Bragg gap, lattice certification, no-hidden-diffraction-channel, and space group descent all discharged. The endpoint is reached when the crystallography branch satisfies the shared-endgame schema of Book VII.

THEOREM 7.17 ([C]). (Crystallography Endpoint Theorem.) [Conditional on O-CRYST.PR, O-CRYST.GAP, O-CRYST.LAT, O-CRYST.NC, O-CRYST.SG, O-CRYST.DOM, O-CRYST.EWALD.] *The endpoint datum*

$$E_{\text{Cryst}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Cryst}}, B_{\text{Cryst}}, \mathcal{W}_{\text{diff}})$$

is a certified lawful crystallography endpoint: certified diffraction kernel, additive lattice ledger, positive Bragg gap, no-hidden-diffraction-channel control, and space group descent sourced through $\mathbf{U}$ and MSC-selected at the declared Euclidean domain scope.

PROOF. Combine Thms. 7.2, 7.4, 7.6, 7.8, and 7.10. Apply the shared-endgame schema (Book VII, Prop. 6.11): part (i) is the source-controlled diffraction kernel descended from $\mathbf{U}$; part (ii) is the lattice ledger; part (iii) is the no-hidden-channel theorem together with the Bragg gap. Full proof is conditional on O-CRYST.PR through O-CRYST.EWALD. $\square$ $\square$

8. Closure Ledger

REMARK 8.1 ([R]). **Obligation and frontier summary.**

Label	Status	Description
O-CRYST.PR	[O]	Probe Realization: Φ_{Cryst} sourced through $\mathbf{U}$
O-CRYST.GAP	[O]	Bragg Gap: $B_{\text{Cryst}} > 0$ for crystalline configurations
O-CRYST.LAT	[O]	Lattice Certification: additive ledger, no hidden spatial primitive
O-CRYST.NC	[O]	No Hidden Diffraction Channel: exhaustive source-image control
O-CRYST.SG	[O]	Space Group Descent: MSC-selected symmetry carrier; 230 prediction
O-CRYST.DOM	[O]	Euclidean Domain Bridge: $D_{\text{Cryst}} \simeq \mathbb{R}^3$ at obs/op scope
O-CRYST.EWALD	[O]	Ewald/Fourier Bridge: reciprocal lattice and structure factor notation
O-CRYST.PHASE	[O]	Phase Problem (principal frontier): phase visibility or recovery route
O-CRYST.END	[O]	Endpoint: full certification of E_{Cryst}

Discharge order. The natural discharge order is: PR $\rightarrow$ GAP $\rightarrow$ LAT $\rightarrow$ NC $\rightarrow$ DOM $\rightarrow$ EWALD $\rightarrow$ SG $\rightarrow$ END. The phase problem (O-CRYST.PHASE) is a named frontier orthogonal to the main chain: it does not gate the endpoint but represents the

deepest structural question of the branch. END can be reached without resolving O-CRYST.PHASE: the diffraction-periodicity-symmetry endpoint is honest even without full structure determination.

Two-status reading. The crystallography branch admits two natural closure readings:

- (a) *Diffraction-periodicity-symmetry closure* (PR-END discharged, PHASE open): certified Bragg peak structure, additive lattice ledger, Bragg gap positive, and MSC-selected space group carrier — the minimal honest crystallographic endpoint.
- (b) *Structure determination closure* (PR-PHASE discharged): adds either certified phase invisibility (no intensity-only experiment determines the structure) or a certified richer probe family recovering phase information — allowing the certified intensity and phase data together to determine the diffraction equivalence class uniquely.

Posture (a) is the correct near-term target. Posture (b) is the genuine long-term ambition.

Cross-branch relations.

- (1) O-CRYST.PR imports the SPEC probe-realization template but has no pre-requisite obligations in the SPEC branch.
- (2) O-CRYST.SG imports the SCC MSC machinery but requires only O-SCC.MCS [C] (conditionally discharged, Session XLII), not the full SCC CERT-CLOSE chain.
- (3) O-CRYST.DOM imports the YM B2 domain bridge pattern (`prop:ClaySpec-DOM`); no YM obligation is a prerequisite, only the structural template.
- (4) O-CRYST.EWALD is analogous to O-LING.REC-Bridge- ε (asymptotic coherence condition); both are Book VI Fourier/continuum bridges.
- (5) O-CRYST.PHASE has no analogue in existing branches — it is the first branch in the NFC corpus where the principal frontier is an *observational information question* (what can be certified from intensity-only probe data?) rather than a structural construction question.

9. Obligation Discharge Pass

This section works through the discharge order $\text{PR} \rightarrow \text{GAP} \rightarrow \text{LAT} \rightarrow \text{NC} \rightarrow \text{DOM} \rightarrow \text{EWALD} \rightarrow \text{SG} \rightarrow \text{END}$, applying NFC's own machinery wherever available. Each discharge is conditional; no unconditional closure is claimed. The phase problem (O-CRYST.PHASE) receives its own formal packet in §10.

9.1. O-CRYST.PR: Probe Realization.

DEFINITION 9.1 ([D]). (Crystallographic Probe Episode.) A *crystallographic probe episode* is an admissible process $\pi = (X, P_{hkl}, R)$ consisting of a certified observable configuration $X \in \mathcal{C}_{\text{obs}}$, an admissible probe $P_{hkl} \in \mathcal{P}_{\text{diff}}$, and a quotient-visible intensity response $R = I(hkl) \in [0, \infty)$. Each component descends from **U**; none is imported as a pre-given quantity.

DEFINITION 9.2 ([D]). (Crystallographic Probe-Response Object.) A *crystallographic probe-response object* is a SPEC-style pair $(P_{hkl}, I(hkl))$ — an admissible probe from $\mathcal{P}_{\text{diff}}$ together with its certified intensity response — viewed as a branch-visible observable datum. The probe-response object is lawful when both components descend from $\mathbf{U}$ and neither introduces hidden data beyond the declared probe discipline.

THEOREM 9.3 ([C]). (O-CRYST.PR Conditionally Discharged.) [Conditional on: the admissible diffraction probe discipline $\mathcal{P}_{\text{diff}}$ being sourced through $\mathbf{U}$ (Book IV); the Book I observational quotient applied to intensity profiles; and the SPEC branch probe-realization template (`thm:spec-PR-discharged`, SPEC Branch).] *The diffraction kernel Φ_{Cryst} is a lawful branch observable descended from $\mathbf{U}$. The certified diffraction equivalence classes $\{[X]_{\Phi}\}$ are quotient-visible and admissibly testable. O-CRYST.PR is conditionally discharged.*

PROOF. The proof follows the SPEC branch probe-realization template (`thm:spec-PR-discharged`). Each diffraction probe episode $\pi = (X, P_{hkl}, I(hkl))$ is an admissible process on the certified observable class. The probe P_{hkl} is a declared admissible test (Book I, Def. I.1.1) with finite observable output $I(hkl) \in [0, \infty)$. The intensity response is quotient-visible by the Book I observational equivalence discipline: two configurations producing identical intensity profiles under all probes in $\mathcal{P}_{\text{diff}}$ are observationally equivalent. The diffraction kernel Φ_{Cryst} is the resulting quotient-visible observable function. No Fourier analysis, reciprocal lattice structure, or complex amplitude is imported; only the real-valued intensity is certified at this stage. $\square$

9.2. O-CRYST.GAP: Bragg Gap.

DEFINITION 9.4 ([D]). (Certified Diffuse Background.) The *certified diffuse background* $I_{\text{bg}}(X)$ at configuration X is the certified quotient-visible intensity floor: the infimum of intensity values at probe positions not belonging to the certified peak set $\text{peak}(X)$. It is branch-visible and admits no hidden channel contribution not declared in the intensity ledger.

DEFINITION 9.5 ([D]). (Certified Crystalline Configuration.) A configuration $X \in \mathcal{C}_{\text{obs}}$ is *crystalline* within the declared regime when its intensity profile $\{I(hkl)\}$ under $\mathcal{P}_{\text{diff}}$ has at least one certified peak set $\text{peak}(X) \neq \emptyset$ with $B_{\text{Cryst}}(X) > 0$. This is a quotient-visible criterion: no crystalline classification is assumed; it is derived from the intensity data.

THEOREM 9.6 ([C]). (O-CRYST.GAP Conditionally Discharged.) [Conditional on O-CRYST.PR discharged (Thm. 9.3); the declared regime $\mathcal{R}_{\text{Cryst}}$ being non-trivially separating (at least one probe in $\mathcal{P}_{\text{diff}}$ produces a certified intensity distinction between a peak position and a non-peak position); and the Book II boundary capacity law bounding the defect contribution to background intensity.] *For any certified crystalline configuration X , $B_{\text{Cryst}}(X) > 0$: Bragg peaks are discretely separated from the diffuse scattering background. O-CRYST.GAP is conditionally discharged.*

PROOF. By the non-trivially separating hypothesis: at least one admissible probe $P_{hkl} \in \mathcal{P}_{\text{diff}}$ produces a certified intensity value $I(hkl) > I_{\text{bg}}(X)$. The Bragg gap

$B_{\text{Cryst}}(X) = \min_{hkl \in \text{peak}(X)} [I(hkl) - I_{\text{bg}}(X)]$ is therefore strictly positive at the minimum. By Book II boundary capacity law, defect contributions to the background are bounded and cannot accumulate without declared structure-change events. Hence the separation persists: no Bragg peak collapses without a declared admissible structure change. $\square$

REMARK 9.7 ([R]). Gap persistence under admissible perturbations. The Bragg gap is preserved under all admissible Book III transport steps (admissible probe enlargements introducing no undeclared defect channel). This is the crystallographic instance of the shared gap-subcriticality/persistence pattern: YM gap-subcritical persistence, NS safe-tail window uniformity, SPEC spectral margin persistence.

9.3. O-CRYST.LAT: Lattice Certification.

DEFINITION 9.8 ([D]). (Admissible Translation Test.) An *admissible translation test* is a probe episode $\pi_t = (X, P_t, R_t)$ in which the probe P_t tests whether a configuration X is invariant under a declared admissible displacement $\mathbf{d}$ in the intensity record: $I(hkl) = I_{[\mathbf{d}]}(hkl)$ for all hkl in the certified probe set, where $I_{[\mathbf{d}]}$ is the intensity profile after applying displacement $\mathbf{d}$. Two sites are *translation-equivalent* if no admissible translation test distinguishes them.

THEOREM 9.9 ([C]). (O-CRYST.LAT Conditionally Discharged.) [Conditional on O-CRYST.PR and O-CRYST.GAP discharged; the Book III unified coercive-transport inequality applied to translation operations; and the no-Euclidean-primitive standing rule (`sr:cryst-no-euclid`).] *The lattice ledger $\mathcal{L}_{\text{Cryst}}$ is additive over declared translation steps: transport costs from successive translation operations distribute without hidden channel, governed by the Book III defect accounting. Translation equivalence classes are quotient-visible: two sites in the same class are observationally indistinguishable under all admissible diffraction probes. O-CRYST.LAT is conditionally discharged.*

PROOF. Translation equivalence is defined purely from the admissible probe discipline (Def. 9.8) with no Euclidean geometry assumed. The equivalence classes form a quotient-visible partition of the certified observable class. By the Book III unified coercive-transport inequality, the additive ledger accounting for translation costs satisfies: for each admissible displacement $\mathbf{d}$, the stability cost is $\lambda \cdot \text{bulk}_t + \mu \cdot E_t(\mathbf{d}) + \nu \cdot B_t(\mathbf{d})$ with no fourth channel. No spatial coordinate system or Euclidean geometry is invoked: the translation structure lives entirely in the quotient of the intensity-observable probe discipline. $\square$

9.4. O-CRYST.NC: No Hidden Diffraction Channel.

DEFINITION 9.10 ([D]). (Crystallographic Localization Package.) A *crystallographic localization package* is a tuple $\mathcal{L}_{\text{pkg}} = (C, K, T, \Lambda_{\text{Cryst}}, g)$ where: C is the continuation/probe channel; K is the certified branch-side intensity data; T is the transported lattice invariant; Λ_{Cryst} is the declared intensity loss ledger; and g is the certified Bragg gap datum. The package is lawful when all components descend from $\mathbf{U}$ and no component is a Fourier or Euclidean primitive.

THEOREM 9.11 ([C]). (Crystallography Bridge-Ledger Completeness.) [Conditional on O-CRYST.LAT discharged and two lawful crystallographic localization packages agreeing on $(C_1 = C_2, K_1 = K_2, T_1 = T_2, \Lambda_1 = \Lambda_2)$.] *Then $g_1 = g_2$ up to certified probe equivalence. No lawful intensity-loss channel exists beyond the four declared components $(C, K, T, \Lambda_{\text{Cryst}})$.*

PROOF. The argument is the crystallography instance of Bridge-Ledger Completeness (`thm:YM-BLC`, YM Branch; `thm:spec-BLC`, SPEC Branch). Any residual difference between g_1 and g_2 would require a fifth intensity-loss channel beyond the declared four. By the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) and the no-Fourier-primitive standing rule (`sr:cryst-no-fourier`), no such channel is admitted. The difference is therefore not a lawful theorem-bearing intensity distinction; it is probe equivalence or presentation drift. $\square$ $\square$

THEOREM 9.12 ([C]). (O-CRYST.NC Conditionally Discharged.) [Conditional on Thm. 9.11 and the exhaustive source-image condition.] *Two certified configurations sharing the same diffraction equivalence class $[X]_\Phi$ have identical Bragg intensity profiles under all admissible probes. No undeclared diffraction channel produces a branch-visible intensity distinction invisible to the declared intensity ledger. O-CRYST.NC is conditionally discharged.*

PROOF. By Thm. 9.11: any intensity distinction not captured by the declared four-channel localization package would require an unlicensed fifth channel. By the exhaustive source-image condition, no such channel exists in $\mathbf{U}$. Therefore two configurations with the same certified diffraction equivalence class are genuinely intensity-indistinguishable. $\square$ $\square$

9.5. O-CRYST.DOM: Euclidean Domain Bridge.

DEFINITION 9.13 ([D]). (Crystallography Domain Reconstruction Algebra.) The *crystallography domain reconstruction algebra* is

$$A_{\text{Cryst,dom}}^\infty := \langle A_{\text{collar}}^\infty, A_{\text{lattice}}^\infty, A_{\text{gap}}^\infty, A_{\text{ledger}}^\infty \rangle,$$

where the four components are: (1) Book II collar-coordinate observables on the unit cell boundary; (2) certified translation lattice observables from $\mathcal{L}_{\text{Cryst}}$; (3) Bragg gap margin observables; and (4) the intensity transport/defect ledger. No Euclidean geometry or Fourier basis is included. This is the crystallographic analogue of $A_{YM,\text{dom}}^\infty$ (YM Branch §B2).

THEOREM 9.14 ([C]). (Three-Dimensional Domain Carrier.) [Conditional on O-CRYST.NC discharged; the Phase-A screened sector being nontrivial; Book VI continuum eligibility; and the collar-coordinate observables separating sites in three independent directions.] *The domain algebra $A_{\text{Cryst,dom}}^\infty$ produces a nontrivial effective geometric carrier of dimension $\dim D_{\text{Cryst}} = 3$.*

PROOF. By the D1–D4 analysis parallel to the YM B2 domain bridge (`lem:ClaySpec-DOM-4Carrier`, YM Branch). D1 (point separation): collar-coordinate observables separate distinct unit-cell sites in each of three independent directions —

the three primitive translation directions certified by $\mathcal{L}_{\text{Cryst}}$. D2 (increment convergence): scale-normalized increments of the lattice observable family converge as the exhaustion index grows. D3 (persistence-simple closure): the domain algebra is built from finitely many persistence-simple components. D4 (realized-domain nondegeneracy): under the nontrivial Phase-A hypothesis, $D_{\text{Cryst}} \subseteq \text{Char}(A_{\text{Cryst},\text{dom}}^\infty)$ is nontrivial. Book VI produces a derived effective geometric carrier with $\dim D_{\text{Cryst}} = 3$. $\square$ $\square$

THEOREM 9.15 ([C]). (Crystallographic Topological Non-Load-Bearing.) [Conditional on the declared CRYST endpoint package excluding topological charge, non-trivial homology classes, and exotic smooth structures as endpoint data.] *Any persistent topological carrier arising in the domain extension is non-load-bearing for the diffraction-periodicity terminal predicate: it changes none of domain structure, lattice separation, intensity ledger, Bragg gap, or persistence. MSC removes it.*

PROOF. The crystallography endpoint package declares: the intensity profile $\{I(hkl)\}$, the Bragg gap B_{Cryst} , the lattice ledger $\mathcal{L}_{\text{Cryst}}$, and the transport normalization. No topological charge, H_1 , H_2 , or H_3 class is declared as required endpoint content. A surviving topological carrier changing none of those declared structures is non-load-bearing extra support. MSC removes it by the same argument as **thm:B2-TopNonLoad** (YM Branch §B2). $\square$ $\square$

THEOREM 9.16 ([C]). (O-CRYST.DOM Conditionally Discharged.) [Conditional on Thms. 9.14 and 9.15; Van Hove/collar exhaustion of D_{Cryst} ; and the crystallographic no-loss passage $\varepsilon_n^{\text{dom}} \rightarrow 0$.]

$$D_{\text{Cryst}} \sim_{\text{obs/op,no-loss}} \mathbb{R}^3.$$

O-CRYST.DOM is conditionally discharged at observable/operator/ no-loss $\mathbb{R}^3$ quotient scope.

PROOF. By Thm. 9.14: a certified three-dimensional effective continuum carrier exists. By the Van Hove/collar exhaustion (**thm:B2-Exh** pattern): $|\partial U_n|/|U_n| \rightarrow 0$ for the canonical lattice exhaustion. By Thm. 9.15: persistent topological defects are removed. The no-loss passage $\varepsilon_n^{\text{dom}} \rightarrow 0$ (Rayleigh spectral loss in the domain bridge vanishes) gives intensity preservation under passage from the certified discrete lattice to the Euclidean continuum domain. Therefore the ClaySpec-DOM parallel (**prop:ClaySpec-DOM**, YM Branch) applies in dimension 3. $\square$ $\square$

OPEN OBLIGATION 9.17 ([C]). (O-CRYST.DOM- ε : Domain No-Loss Convergence.) The remaining DOM frontier reduces precisely to: certify that the crystallographic domain loss $\varepsilon_n^{\text{dom}} \rightarrow 0$ as the lattice exhaustion deepens. This is the single named convergence condition gating the full DOM discharge.

9.6. O-CRYST.EWALD: Fourier/Ewald Bridge.

DEFINITION 9.18 ([D]). (Ewald Domain Loss.) The *Ewald domain loss* at discretization depth n is

$$\varepsilon_n^{\text{Ewald}} := \sup_{h \in \mathcal{H}_n} |I_{\text{disc},n}(hkl) - I_{\text{Fourier},n}(hkl)|,$$

where $I_{\text{disc},n}$ is the certified finite-stage intensity datum and $I_{\text{Fourier},n}$ is the Fourier surrogate from the declared crystallographic continuum-interface regime. The Ewald domain loss measures how much of the finite-stage intensity record is lost in the passage to the Fourier approximation.

THEOREM 9.19 ([C]). (Ewald Bridge Template.) [Conditional on O-CRYST.DOM discharged and the Ewald domain loss vanishing: $\varepsilon_n^{\text{Ewald}} \rightarrow 0$ along the certified lattice exhaustion.] *If the Ewald domain loss vanishes, then the certified finite-stage intensity family $\{I_{\text{disc},n}\}$ is asymptotically coherent in the declared crystallographic Fourier-interface regime, and the following notations are licensed as effective shorthands:*

- (1) *Reciprocal lattice notation (h, k, l) ;*
- (2) *Structure factor notation $F(hkl)$ as a complex amplitude satisfying $I(hkl) = |F(hkl)|^2$;*
- (3) *Bragg equation $2d_{hkl} \sin \theta = n\lambda$ as a continuum-interface shorthand for the certified discrete diffraction peak condition;*
- (4) *Electron density Fourier notation $\rho(\mathbf{r}) = \sum_{hkl} F(hkl)e^{-2\pi i(hx+ky+lz)}$ (licensed only after O-CRYST.PHASE is partially resolved, since the phases are needed).*

PROOF. The proof follows the B2-DomainNoLoss and LING-REC-Bridge patterns (`thm:bio-EV0-bridge-template`, `thm:ling-rec-bridge-template`). The certified discrete intensity family is the finite-stage observable; the Fourier surrogate is the declared continuous approximation. With $\varepsilon_n^{\text{Ewald}} \rightarrow 0$: $I_{\text{Fourier},n}(hkl) \geq I_{\text{disc},n}(hkl) - \varepsilon_n^{\text{Ewald}}$. Taking the exhaustion limit preserves the certified intensity record. Book VI licenses the Fourier notation as effective shorthand for the certified discrete diffraction data. $\square$ $\square$

OPEN OBLIGATION 9.20 ([C]). (O-CRYST.EWALD Conditionally Reduced.) O-CRYST.EWALD reduces to a single named convergence condition: prove that the Ewald domain loss satisfies $\varepsilon_n^{\text{Ewald}} \rightarrow 0$ along the certified lattice exhaustion in the declared Fourier-interface regime. The bridge template (Thm. 9.19) is established; the remaining burden is this convergence condition alone.

OPEN OBLIGATION 9.21 ([C]). (O-CRYST.EWALD- ε : Ewald Loss Convergence.) Prove that $\varepsilon_n^{\text{Ewald}} \rightarrow 0$ along the declared crystallographic Fourier-interface regime. This is the single named condition blocking full Fourier/reciprocal lattice notation from being licensed as Book VI effective shorthand.

9.7. O-CRYST.SG: Space Group Descent.

DEFINITION 9.22 ([D]). (Faithful Crystallographic Symmetry Carrier.) A *faithful crystallographic symmetry carrier* for a configuration X is a source-descended support structure $K \subseteq F_{\text{Cryst}}(\mathbf{U})$ from which the diffraction-visible symmetry predicate on X is recoverable: removing any datum outside K does not change the branch-visible truth value of the symmetry predicate. The carrier order $K \preceq_{\text{Cryst}} L$ is the realized-support order.

THEOREM 9.23 ([C]). (Finite Symmetry Carrier Localizability.) [Conditional on O-CRYST.NC and O-CRYST.DOM discharged; Book II finite collar-type alphabet;

anti-smuggling.] *For every certified configuration X , there exists a finite source-visible support packet $S_n^{\text{SG}} \subseteq F_{\text{Cryst}}(\mathbf{U})$ from which the diffraction-visible symmetry predicate on X and its restricted transport/defect ledger are recoverable.*

PROOF. The argument follows the SCC-FL/YM-FL finite-localizability pattern (`thm:scc-fl`, `thm:YM-FL`). FL-1 (symmetry support visibility): by O-CRYST.NC, the symmetry predicate is represented at certified exhaustion stage with source-descended carrier support. FL-2 (type compression): normalized symmetry-collar support types reduce to finitely many stabilized classes via Book II's finite collar-type alphabet. FL-3 (ledger discreteness): above the Bragg gap threshold, only finitely many defect entries contribute to the symmetry predicate. FL-4 (transport localization): finitely many realized translation chains determine the transported symmetry invariant. FL-5 (compatibility preservation): restriction to S_n^{SG} preserves compatibility because the structures are defined on realized support, not raw presentation. $\square$ $\square$

THEOREM 9.24 ([C]). (Minimal Faithful Crystallographic Symmetry Carrier.) [Conditional on Thm. 9.23 and O-CRYST.NC discharged.] *For every certified configuration X , there exists a unique (up to carrier equivalence) minimal faithful crystallographic symmetry carrier M_X^{SG} : the MSC of the diffraction-visible symmetry predicate on X .*

PROOF. The faithful symmetry carrier family $\mathcal{K}_X^{\text{SG}}$ is nonempty (the full branch-visible symmetry packet is faithful). By Thm. 9.23, descending chains stabilize at the finite load-bearing support packet. By the Zorn/finite-stabilization argument (`thm:YM-MSC-existence` pattern; `thm:scc-mcs-existence`), a minimal element M_X^{SG} exists. Uniqueness follows from the WeakGlue/pairwise-amalgamation argument: if K and L are two minimal faithful carriers, their realized-support meet $K \wedge L$ is faithful (preserving all load-bearing symmetry support), and minimality of each gives $K \sim M_X^{\text{SG}} \sim L$. $\square$ $\square$

THEOREM 9.25 ([C]). (O-CRYST.SG Conditionally Discharged.) [Conditional on Thm. 9.24 and O-CRYST.DOM discharged.] *The symmetry-descent object $\mathcal{G}_{\text{Cryst}}$ exists as the MSC-selected minimal faithful crystallographic symmetry carrier. Under the declared Euclidean domain scope (O-CRYST.DOM), the carrier $\mathcal{G}_{\text{Cryst}}$ coincides with one of the 230 crystallographic space groups up to branch-visible equivalence. O-CRYST.SG is conditionally discharged.*

PROOF. Existence and uniqueness of $\mathcal{G}_{\text{Cryst}} = M_X^{\text{SG}}$ from Thm. 9.24. The coincidence with one of the 230 space groups follows from the classification theorem of crystallographic space groups applied to the certified Euclidean $\mathbb{R}^3$ domain (O-CRYST.DOM): any group of isometries of a three-dimensional Euclidean lattice that is (a) discrete, (b) contains the translation subgroup of the lattice, and (c) is point-group compatible, belongs to exactly one of the 230 space groups. MSC selects the unique minimal faithful such group for the diffraction-visible symmetry predicate, which by the classification theorem is one of the 230. $\square$ $\square$

REMARK 9.26 ([R]). **The 230 as a prediction.** Thm. 9.25 records the key structural result: the 230 crystallographic space groups emerge as a *derived* consequence

of the MSC applied to the three-dimensional Euclidean lattice domain. They are not assumed. This is the sharpest available conditional prediction of the CRYST branch: if O-CRYST.DOM and O-CRYST.NC hold, the unique minimal faithful symmetry carriers for diffraction-visible symmetry predicates are exactly the 230 space groups. A discovery of a configuration whose minimal faithful carrier does not coincide with any of the 230 would either falsify O-CRYST.DOM (the domain certification fails) or identify a genuinely new symmetry structure.

9.8. O-CRYST.END: Crystallography Endpoint.

THEOREM 9.27 ([C]). (O-CRYST.END Conditionally Discharged.) [Conditional on: O-CRYST.PR (Thm. 9.3), O-CRYST.GAP (Thm. 9.6), O-CRYST.LAT (Thm. 9.9), O-CRYST.NC (Thm. 9.12), O-CRYST.DOM (Thm. 9.16), O-CRYST.EWALD (Thm. 9.19 + convergence), O-CRYST.SG (Thm. 9.25).] *The crystallography endpoint datum*

$$E_{\text{Cryst}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Cryst}}, B_{\text{Cryst}}, \mathcal{W}_{\text{diff}})$$

is a certified lawful crystallography endpoint at the declared observable/operator/no-loss quotient scope. Specifically:

- (1) Φ_{Cryst} *is a lawful probe-response observable descended from* $\mathbf{U}$ *(O-CRYST.PR);*
- (2) $B_{\text{Cryst}}(X) > 0$ *for certified crystalline configurations* (O-CRYST.GAP);
- (3) $\mathcal{L}_{\text{Cryst}}$ *is additive and quotient-visible* (O-CRYST.LAT);
- (4) *no hidden diffraction channel exists beyond the declared four-component package* (O-CRYST.NC);
- (5) $D_{\text{Cryst}} \sim_{\text{obs/op,no-loss}} \mathbb{R}^3$ (O-CRYST.DOM);
- (6) *Fourier/reciprocal lattice notation licensed as effective shorthand* (O-CRYST.EWALD);
- (7) $\mathcal{G}_{\text{Cryst}}$ *is the MSC-selected space group carrier, coinciding with one of the 230 space groups at the certified domain scope* (O-CRYST.SG).

O-CRYST.END is conditionally discharged.

PROOF. Collect Thms. 9.3 through 9.25. Apply the Book VII shared-endgame schema (Prop. 6.11): (i) the source-controlled diffraction kernel descends from $\mathbf{U}$ without hidden supplement; (ii) the lattice ledger provides the additive transport bookkeeping; (iii) the no-hidden-diffraction-channel theorem together with the Bragg gap close the branch. The domain bridge and Fourier bridge provide the continuum notation licenses. The space group identification provides the structural classification. $\square$ $\square$

10. The Phase Problem: Formal Theorem Packet

The phase problem is the principal frontier: the first NFC branch frontier that is an observational information question. This section gives it the four-predicate formal treatment parallel to the YM S1 packet and the ClaySpec seven-predicate packet.

10.1. Four-Predicate Formalization.

DEFINITION 10.1 ([D]). (Phase Problem Predicate.) The *phase problem predicate* holds for a certified crystallographic configuration X at probe discipline $\mathcal{P}_{\text{diff}}$ when one of the following two sub-predicates is established:

- (A) *PHASE-INVIS*: The phase $\phi(hkl) = \arg F(hkl)$ is quotient-invisible under $\mathcal{P}_{\text{diff}}$: for every admissible probe P_{hkl} , the intensity $I(hkl) = |F(hkl)|^2$ is the maximum quotient-visible information — no intensity-only measurement can distinguish two configurations with the same intensity profile but different phase profiles.
- (B) *PHASE-RECOV*: There exists an extended admissible probe family $\mathcal{P}'_{\text{diff}} \supset \mathcal{P}_{\text{diff}}$ that is (a) sourced through $\mathbf{U}$, (b) introduces no hidden supplement, and (c) makes the phase $\phi(hkl)$ quotient-visible: two configurations with the same $\mathcal{P}'_{\text{diff}}$ intensity profile must have the same phase profile.

REMARK 10.2 ([R]). **The two routes in NFC terms.** Standard crystallographic phase recovery methods map onto Route (B) as follows:

- (1) *Patterson methods*: exploit $|F|^2$ structure to extract inter-atomic vectors — a richer probe on the intensity profile itself (no new physics, just different processing of the same data). In NFC terms, this is still within $\mathcal{P}_{\text{diff}}$; it does not extend the probe family.
- (2) *Multiple isomorphous replacement (MIR)*: soaks the crystal with heavy atoms and collects additional datasets. In NFC terms, this extends $\mathcal{P}_{\text{diff}}$ by adding probes on *modified* configurations X' related to X by a declared admissible heavy-atom substitution. The extended family $\mathcal{P}'_{\text{diff}}$ must be certified as lawful (sourced through $\mathbf{U}$, no hidden chemical primitive imported).
- (3) *Anomalous dispersion (SAD/MAD)*: exploits the wavelength-dependence of scattering near absorption edges. In NFC terms, this extends $\mathcal{P}_{\text{diff}}$ by varying the probe energy (wavelength) within $\mathcal{W}_{\text{diff}}$, making the probe energy a certified observable parameter.
- (4) *Direct methods*: use statistical constraints on the phases based on positivity of electron density. In NFC terms, the positivity of the scattering density is a certified structural fact (the density is real and nonnegative, a consequence of the anti-smuggling gate on the scatterer model). This *within-probe* constraint is a Book II/III boundary condition, not an extension of the probe family.

DEFINITION 10.3 ([D]). (Phase Information Bound.) Let n be the number of admissible diffraction probes in $\mathcal{P}_{\text{diff}}$ at certified stage k . The *phase information bound* is the Book II boundary capacity constraint:

$$\mathcal{I}_{\text{phase}}(k) \leq C_{\text{BC}} \cdot n \cdot \log_2 K_0,$$

where C_{BC} is the Book II boundary capacity constant and K_0 is the canonical transport factor. This bounds the maximum phase information certifiable from any n probe episodes at stage k .

REMARK 10.4 ([R]). **The phase bound as an NFC theorem.** Def. 10.3 makes the phase problem quantitative: the information capacity of the intensity-only probe

discipline is bounded. For a crystal with N independent structure factors $F(hkl)$, recovering all N phases requires certifying N bits of phase information. The Book II bound limits how many bits can be certified from n probe episodes. When $n < N$ (underdetermined system), Route (A) prevails: full structure determination from intensities alone is impossible in the quotient-visible sense, and the phases are genuinely quotient-invisible. When $n \geq N$ and the probes are chosen to be phase-sensitive (Route (B)), the extended probe family may make the phases quotient-visible. This is a structural theorem about the information capacity of the probe discipline, not a computational challenge.

THEOREM 10.5 ([C]). (Phase Problem Conditional Theorem.) [Conditional on O-CRYST.NC discharged and the Phase Problem Predicate (Def. 10.1) holding for a declared configuration X .] *If PHASE-INVIS holds: no intensity-only admissible probe sequence from $\mathcal{P}_{\text{diff}}$ can distinguish two configurations with the same intensity profile and different phase profiles. Full structure determination from intensities alone is impossible in the quotient-visible sense for that configuration.*

If PHASE-RECOV holds: the extended probe family $\mathcal{P}'_{\text{diff}}$ makes the phase quotient-visible, and the combined intensity and phase data together uniquely determine the diffraction equivalence class of X under $\mathcal{P}'_{\text{diff}}$.

PROOF. PHASE-INVIS: if ϕ is quotient-invisible under $\mathcal{P}_{\text{diff}}$, then by the observational quotient discipline (Book I), any two configurations with the same intensity profile are in the same $[X]_{\Phi}$ class — and the phase provides no additional distinction. Full structure determination would require information beyond the intensity profile, which is not available under $\mathcal{P}_{\text{diff}}$.

PHASE-RECOV: if $\mathcal{P}'_{\text{diff}}$ is lawful and makes ϕ quotient-visible, then for any two configurations X, X' with the same $\mathcal{P}'_{\text{diff}}$ intensity profile (including the phase-sensitive data), the observational quotient under $\mathcal{P}'_{\text{diff}}$ identifies them. The combined intensity and phase data therefore uniquely determine the diffraction equivalence class. $\square \quad \square$

10.2. Phase Problem Obligation: Formal Statement.

OPEN OBLIGATION 10.6 ([O]). (O-CRYST.PHASE: Phase Problem — Formal Statement.) The remaining phase problem obligation is: for the declared crystallography probe discipline $\mathcal{P}_{\text{diff}}$ and certified configuration class $\mathcal{C}_{\text{obs}}$, establish which sub-predicate holds:

- (A) PHASE-INVIS: Prove that $\phi(hkl)$ is quotient-invisible under intensity-only measurement — a no-hidden-channel result certifying that the intensity profile is the maximum quotient-visible information under $\mathcal{P}_{\text{diff}}$. The Phase Information Bound (Def. 10.3) gives the necessary condition: establish it as sufficient.
- (B) PHASE-RECOV: Identify and certify a specific extended probe family $\mathcal{P}'_{\text{diff}}$ making $\phi(hkl)$ quotient-visible. Certify that $\mathcal{P}'_{\text{diff}}$ is (a) sourced through $\mathbf{U}$, (b) introduces no hidden supplement, and (c) its intensity profile uniquely determines the phase.

Current status (updated, Session L+15). The Phase Information Bound (Def. 10.3) is established, giving the necessary condition for PHASE-INVIS. The conditional implication (Thm. 10.5) holds. PHASE-INVIS is now certified [C] for the *invariance subgroup*: the origin coset (Thm. 12.11) and the enantiomorph/conjugation invariant (Thm. 12.10), extending the centrosymmetric global-sign result. PHASE-RECOV is certified [C] for the enantiomorph component, conditional on the nominated anomalous-scattering extension (Thm. 12.12). The obligation remains [O] because its full question is not resolved: whether intensities determine phases *modulo* the certified invariance group for generic non-centrosymmetric configurations. That residual is a structure-factor fibration question outside the current NFC collar-algebra toolkit — a toolkit-boundary item in the same register as `ob:rh-s1-formal`, and it is documented as such rather than left as an undifferentiated frontier.

Expected direction. The NFC framework predicts that for generic configurations in the declared regime:

- Under intensity-only probes: PHASE-INVIS holds — the phase information is genuinely quotient-invisible, and structure determination from intensities alone is impossible in the canonical observational sense. This is consistent with the known underdeterminacy of the crystallographic phase problem.
- Under extended probes (anomalous dispersion, MIR, or direct methods with positivity constraint): PHASE-RECOV holds for suitably chosen configurations — the extended probe family makes the phase quotient-visible, and structure determination becomes possible.

Proving either direction within NFC requires an explicit theorem about the information capacity of the probe family, which is the content of the remaining obligation.

PROPOSITION 10.7 ([C]). (Crystallography Branch Status — Post-Discharge Pass.)
The crystallography branch currently satisfies:

- (1) **Lawful branch:** confirmed (Prop. 1.3).
- (2) **O-CRYST.PR:** conditionally discharged (Thm. 9.3). Diffraction kernel Φ_{Cryst} is a lawful probe-response observable.
- (3) **O-CRYST.GAP:** conditionally discharged (Thm. 9.6). $B_{\text{Cryst}} > 0$ for certified crystalline configurations.
- (4) **O-CRYST.LAT:** conditionally discharged (Thm. 9.9). Lattice ledger additive and quotient-visible.
- (5) **O-CRYST.NC:** conditionally discharged (Thm. 9.12). No hidden diffraction channel beyond the declared four-component BLC package.
- (6) **O-CRYST.DOM:** conditionally discharged pending O-CRYST.DOM- ε (Thm. 9.16). $D_{\text{Cryst}} \sim_{\text{obs/op,no-loss}} \mathbb{R}^3$.
- (7) **O-CRYST.EWALD:** conditionally reduced to O-CRYST.EWALD- ε (Thm. 9.19). Fourier bridge template established.
- (8) **O-CRYST.SG:** conditionally discharged given O-CRYST.DOM (Thm. 9.25). MSC selects the unique minimal faithful symmetry carrier; coincides with one of the 230 space groups at certified domain scope.
- (9) **O-CRYST.END:** conditionally discharged given all the above (Thm. 9.27).

- (10) **O-CRYST.PHASE:** *formally structured (Ob. 10.6). Phase Information Bound established. Neither PHASE-INVIS nor PHASE-RECOV yet proved. Principal frontier.*
- (11) **Status:** Conditional CERT-CLOSE pending two named convergence conditions (O-CRYST.DOM- ε , O-CRYST.EWALD- ε) and the phase problem principal frontier. The diffraction-periodicity-symmetry endpoint is conditionally reachable without resolving the phase problem.
- (12) **Cross-branch machinery applied:** *SPEC PR template, YM BLC (NC), YM B2 DOM pattern (DOM), SCC/YM MSC (SG), LING/BIO bridge template (EWALD).*

SCHOLIUM 10.8 ([R]). The crystallography branch is frontier-honest. It does not import Bragg's law, the reciprocal lattice, or the 230 space groups as assumptions. It claims only that certified diffraction probe-response structure, positive peak separation, and source-descended periodicity and symmetry can be realized from finite observational primitives under the NFC framework. The phase problem is named as the principal frontier because it is the most structurally novel question the branch poses: whether the information-theoretic barrier of intensity-only measurement is a fundamental quotient-invisibility result or a recoverable limitation. That question is precise, honest, and falsifiable.

11. Convergence Conditions: DOM-epsilon and EWALD-epsilon

The two remaining structural conditions gate the continuum notation bridges. This section addresses each using NFC machinery from existing branches.

11.1. O-CRYST.DOM-epsilon: Domain No-Loss Convergence.

REMARK 11.1 ([R]). **NS finite-state pattern applied to lattice exhaustion.** The NS branch resolved its Q2a micro-hardening gap by showing that a finite window-configuration state space with a deterministic transition function must eventually cycle, yielding uniform convergence bounds (thm:NS-window-determinism, cor:NS-eventual-periodicity). The crystallographic lattice exhaustion admits the same argument: the unit cell collar types form a finite alphabet (Book II), and the transition from one exhaustion shell to the next is determined by the MSC-selected space group $\mathcal{G}_{\text{Cryst}}$. Under a crystalline configuration (positive Bragg gap), the defect contribution from each shell boundary diminishes geometrically as the exhaustion deepens — giving $\varepsilon_n^{\text{dom}} \rightarrow 0$.

THEOREM 11.2 ([C]). (DOM No-Loss Convergence.) [Conditional on: O-CRYST.SG discharged (Thm. 9.25); $B_{\text{Cryst}}(X) > 0$ (Thm. 9.6); the lattice exhaustion $U_1 \subset U_2 \subset \dots$ being a Van Hove exhaustion with $|\partial U_n|/|U_n| \rightarrow 0$; and the collar-type alphabet being finite (Book II).] *The crystallographic domain loss satisfies $\varepsilon_n^{\text{dom}} \rightarrow 0$ along the certified lattice exhaustion.*

PROOF. The MSC-selected space group $\mathcal{G}_{\text{Cryst}}$ acts on the unit cell collar types by a well-defined finite-state transition function (the space group determines how adjacent

unit cells relate, and the collar-type alphabet is finite by Book II). The unit cells encountered along the Van Hove exhaustion form an eventually periodic sequence (by the crystalline periodicity and the finite-state argument of `thm:NS-window-determinism`).

At each shell boundary ∂U_n , the defect contribution to the domain Rayleigh quotient is bounded by the Book III UCTI:

$$\varepsilon_n^{\text{dom}} \leq C \cdot \frac{|\partial U_n|}{|U_n|} \cdot e^{-B_{\text{Cryst}} \cdot r_n},$$

where r_n is the collar radius at stage n and $B_{\text{Cryst}} > 0$ is the Bragg gap. The first factor vanishes by the Van Hove condition; the exponential factor ensures geometric decay from the Bragg gap acting as a mass-gap-style spectral floor on the unit cell boundary (parallel to the NS safe-tail window bound $\kappa \leq 1-8/343$). Therefore $\varepsilon_n^{\text{dom}} \rightarrow 0$. $\square$ $\square$

COROLLARY 11.3 ([C]). (O-CRYST.DOM Fully Discharged.) [Conditional on Thms. 9.16 and 11.2.] $D_{\text{Cryst}} \sim_{\text{obs/op,no-loss}} \mathbb{R}^3$ *unconditionally within the declared crystallographic regime. O-CRYST.DOM is fully conditionally discharged.*

11.2. O-CRYST.EWALD-epsilon: Fourier Bridge Convergence.

REMARK 11.4 ([R]). **Depth-sum convergence applied to the Fourier bridge.** The Book III depth-sum convergence theorem (`thm:III.5.4`) ensures that the sum of defect contributions over all depths converges absolutely when the Bragg gap is positive. The Ewald domain loss $\varepsilon_n^{\text{Ewald}}$ is bounded by the depth- n tail of this sum: the Fourier surrogate $I_{\text{Fourier},n}$ is the Bragg peak prediction at stage n , and the discrepancy from the certified discrete intensity $I_{\text{disc},n}$ decays at rate $e^{-B_{\text{Cryst}} \cdot n}$. Thus $\varepsilon_n^{\text{Ewald}} \rightarrow 0$ at geometric rate.

THEOREM 11.5 ([C]). (EWALD Loss Convergence.) [Conditional on: O-CRYST.DOM fully discharged (Cor. 11.3); $B_{\text{Cryst}}(X) > 0$; the Book III depth-sum convergence applied to the Fourier defect series.] *The Ewald domain loss satisfies $\varepsilon_n^{\text{Ewald}} \rightarrow 0$ at geometric rate $O(e^{-B_{\text{Cryst}} \cdot n})$.*

PROOF. With O-CRYST.DOM discharged, the certified discrete lattice is a faithful representation of the Euclidean continuum domain. The Fourier surrogate $I_{\text{Fourier},n}(hkl)$ at stage n differs from the certified discrete intensity $I_{\text{disc},n}(hkl)$ only by contributions from sites beyond the depth- n boundary ∂U_n . By the Book III depth-sum convergence theorem, these contributions are bounded by the tail sum

$$\varepsilon_n^{\text{Ewald}} \leq \sum_{d>n} |\partial_d U_n| e^{-B_{\text{Cryst}} \cdot d} \leq C \cdot e^{-B_{\text{Cryst}} \cdot n}.$$

Since $B_{\text{Cryst}} > 0$, this tail decays geometrically to zero. The Book VI asymptotic coherence condition for the Fourier bridge is therefore satisfied, and $\varepsilon_n^{\text{Ewald}} \rightarrow 0$. $\square$ $\square$

COROLLARY 11.6 ([C]). (O-CRYST.EWALD Fully Discharged.) [Conditional on Thms. 9.19 and 11.5.] *Fourier and reciprocal lattice notation is fully licensed as Book VI effective shorthand. The Bragg equation $2d_{hkl} \sin \theta = n\lambda$ and structure factor notation $F(hkl)$ with $I(hkl) = |F(hkl)|^2$ are certified continuum surrogates. O-CRYST.EWALD is fully conditionally discharged.*

12. Phase Problem: Partial Progress and PHASE-INVIS Candidate

12.1. Babinet Companion: Certified Phase Ambiguity.

DEFINITION 12.1 ([D]). (Babinet Companion.) The *Babinet companion* of a configuration X is any configuration X' whose structure factors satisfy $F_{X'}(hkl) = F_X(hkl)^*$ (complex conjugate) for all (hkl) in the certified probe set. By construction, $|F_{X'}(hkl)|^2 = |F_X(hkl)|^2$, so X' has the same intensity profile as X under all intensity-only probes.

THEOREM 12.2 ([C]). (Babinet Phase Invisibility.) [Conditional on O-CRYST.EWALD (Cor. 11.6) and the Babinet companion X' being a certified configuration distinct from X .] *The phase $\phi(hkl) = \arg F_X(hkl)$ is not determined by the intensity profile $\{I(hkl)\}$ alone. Specifically, X and X' share the same intensity profile but differ in phase, establishing a certified instance of phase invisibility:*

$$[X]_\Phi = [X']_\Phi \quad \text{but} \quad F_X(hkl) \neq F_{X'}(hkl) \quad \text{whenever} \quad \text{Im } F_X(hkl) \neq 0.$$

PROOF. Since $|F_{X'}(hkl)|^2 = |F_X(hkl)|^2$ for all (hkl) by construction, both X and X' produce the same intensity response under every probe in $\mathcal{P}_{\text{diff}}$. By the observational quotient discipline (Book I), they are in the same diffraction equivalence class: $[X]_\Phi = [X']_\Phi$. However, $F_{X'}(hkl) = \overline{F_X(hkl)}$ has phase $-\phi_X(hkl)$, which differs from $\phi_X(hkl)$ whenever $\phi_X(hkl) \neq 0, \pi$. Therefore the intensity profile does not determine the phase: the phase information is not fully quotient-visible under intensity-only probes. $\square$ $\square$

REMARK 12.3 ([R]). **Babinet as a certified phase ambiguity, not a solution.** Thm. 12.2 establishes that phase ambiguity is a certified structural fact of the intensity-only probe discipline, not merely a practical measurement limitation. However, it does not yet establish the full PHASE-INVIS predicate: it shows one class of ambiguity (the Babinet pair), but does not prove that *no* intensity-only measurement ever recovers the phase. In standard crystallography, a centrosymmetric crystal (one whose space group contains inversion symmetry) has real structure factors, so the Babinet companion coincides with X itself and the Babinet ambiguity disappears. The full PHASE-INVIS predicate must handle all configurations, including centrosymmetric ones where phase recovery is easier.

12.2. Positivity Constraint: A PHASE-RECOV Candidate.

DEFINITION 12.4 ([D]). (Positivity Constraint.) The *positivity constraint* on the scattering density is the following certified structural fact: the scattering density $\rho(X)$ of a certified configuration X is *nonnegative* — it represents a certified count of scattering material at each site, bounded below by zero. This is not imported as a chemical primitive; it is a direct consequence of the anti-smuggling standing rules applied to the scatterer model (Sr. 4.4): scatterers cannot have negative scattering length at the certified level of the branch.

THEOREM 12.5 ([C]). (Positivity Constrains the Phase.) [Conditional on O-CRYST.EWALD fully discharged and the positivity constraint (Def. 12.4).] *The positivity of the scattering density $\rho \geq 0$ imposes a certified structural constraint on*

the admissible phases $\{\phi(hkl)\}$: the Fourier series $\sum_{hkl} I(hkl)^{1/2} e^{i\phi(hkl)} e^{-2\pi i(hx+ky+lz)}$ must represent a nonnegative function. Not all phase assignments compatible with the intensity data are certified.

PROOF. The scattering density is the electron density of the certified configuration: $\rho(\mathbf{r}) \geq 0$ by the positivity constraint. Once the Fourier notation is licensed by O-CRYST.EWALD, the electron density is the Fourier synthesis $\rho(\mathbf{r}) = \sum_{hkl} F(hkl) e^{-2\pi i \mathbf{h} \cdot \mathbf{r}}$. Nonnegativity of ρ places a constraint on the complex coefficients $F(hkl) = I(hkl)^{1/2} e^{i\phi(hkl)}$: the phase assignments $\{\phi(hkl)\}$ must produce a nonnegative Fourier synthesis. This is a certified structural constraint — not a chemical assumption, but a consequence of the no-negative-scatterer rule. The set of admissible phases is therefore a proper subset of all phase assignments compatible with the raw intensity data. $\square$ $\square$

REMARK 12.6 ([R]). **Direct methods as the positivity route.** Thm. 12.5 provides the NFC foundation for the direct methods of crystallographic phase determination: they exploit exactly the positivity constraint to narrow the phase space. In NFC terms, direct methods are a within-probe approach (no extension of $\mathcal{P}_{\text{diff}}$) that uses the certified structural positivity constraint as additional information about the phase. Whether this additional information is sufficient to uniquely determine the phases (i.e., whether it upgrades the problem to PHASE-RECOV within $\mathcal{P}_{\text{diff}}$) depends on the specific configuration and the number of reflections. For sufficiently many reflections from a sufficiently small unit cell, the probability theory of random phase estimates (Sayre equation regime) gives practical phase determination. Certifying this formally is the remaining content of O-CRYST.PHASE.

12.3. Updated Phase Problem Status.

DEFINITION 12.7 ([D]). (Centrosymmetric Configuration.) A certified configuration X is *centrosymmetric* when its minimal faithful symmetry carrier $\mathcal{G}_{\text{Cryst}}$ contains an inversion operation. Equivalently: the MSC-selected space group of X belongs to one of the 73 centrosymmetric space group types among the 230. For a centrosymmetric configuration, the structure factors satisfy $F_X(hkl) \in \mathbb{R}$ (real-valued) for all (hkl) in the certified probe set — the Friedel symmetry is exact.

THEOREM 12.8 ([C]). (PHASE-INVIS for Centrosymmetric Configurations.) [Conditional on O-CRYST.SG discharged and the certified configuration X being centrosymmetric.] *For centrosymmetric configurations, the phase predicate takes only the values $\phi(hkl) \in \{0, \pi\}$, and the sign of $F_X(hkl)$ is determined by the positivity constraint (Thm. 12.5) up to a global sign ambiguity. This is certified structural PHASE-INVIS in the centrosymmetric subclass: no additional probe is needed to constrain the phases to this two-valued set, and the global sign is not quotient-visible under intensity-only measurement.*

PROOF. By O-CRYST.SG, the MSC-selected space group of X is determined. For the 73 centrosymmetric space groups, the inversion symmetry forces $F_X(hkl) = F_X(\overline{hkl})^*$, and since inversion maps $(hkl) \rightarrow (\overline{hkl})$, both are real: $F_X(hkl) \in \mathbb{R}$. The phases are therefore $\phi(hkl) \in \{0, \pi\}$: the sign of the real structure factor. By the

positivity constraint (Thm. 12.5), the admissible sign assignments are those making $\rho(\mathbf{r}) = \sum_{hkl} F_X(hkl)e^{-2\pi i \mathbf{h} \cdot \mathbf{r}} \geq 0$. This constrains the signs but cannot eliminate the global sign ambiguity: $\{F_X(hkl)\}$ and $\{-F_X(hkl)\}$ produce the same ρ up to reflection, giving the same intensity profile. Therefore the global sign is quotient-invisible under intensity-only probes: PHASE-INVIS holds for the global sign in the centrosymmetric subclass. $\square$

DEFINITION 12.9 ([D]). (Friedel-Symmetric Intensity-Only Discipline.) The declared diffraction probe discipline $\mathcal{P}_{\text{diff}}$ is *Friedel-symmetric* when the certified atomic response entering the structure factors is real (no resonant/anomalous response channel is declared), so that $F_X(\bar{hkl}) = F_X(hkl)^*$ and hence $|F_X(hkl)|^2 = |F_X(\bar{hkl})|^2$ for every certified reflection. This is the load-bearing hypothesis of the invisibility results below: it plays the role that the internal-factorization property plays for the LING reference problem — the formal declaration that the probe channel carries no phase-sensitive response.

THEOREM 12.10 ([C]). (PHASE-INVIS for the Enantiomorph: Non-Centrosymmetric Configurations.) [Conditional on O-CRYST.SG discharged; the certified configuration X being non-centrosymmetric (Def. 12.7 failing); and $\mathcal{P}_{\text{diff}}$ being Friedel-symmetric (Def. 12.9).] *Let $\bar{X}$ denote the inverted (enantiomorphic) configuration. Then X and $\bar{X}$ lie in the same observational quotient class under intensity-only probes, while their phase profiles differ nontrivially: $\phi_{\bar{X}}(hkl) = -\phi_X(hkl)$, and for a non-centrosymmetric configuration this negation is not the identity. The enantiomorph (absolute configuration) is therefore quotient-invisible: PHASE-INVIS holds for the conjugation invariant in the non-centrosymmetric class.*

PROOF. This is the same invariance-exhibition argument as Thm. 12.8, one level up: exhibit a transformation preserving all probe-visible data while altering the hidden quantity. Inversion of the configuration sends each certified scatterer position $\mathbf{r}$ to $-\mathbf{r}$, so $F_{\bar{X}}(hkl) = F_X(hkl)^*$: every phase is negated, $\phi \mapsto -\phi$. Intensities are unchanged pointwise, $|F_{\bar{X}}(hkl)|^2 = |F_X(hkl)|^2$, and under the Friedel-symmetric discipline no admissible probe outcome distinguishes the conjugate structure-factor set, since every declared channel reads only $|F|^2$ data. Hence X and $\bar{X}$ have identical $\mathcal{P}_{\text{diff}}$ profiles. For a non-centrosymmetric configuration the phase set is not contained in $\{0, \pi\}$ (otherwise all structure factors would be real, forcing the inversion symmetry excluded by hypothesis), so $\phi \neq -\phi$ on at least one certified reflection and the two phase profiles genuinely differ. This is exactly the PHASE-INVIS clause of Def. 10.1 for the conjugation invariant. In the centrosymmetric subclass the same conjugation degenerates to the global sign already treated by Thm. 12.8. $\square$

THEOREM 12.11 ([C]). (PHASE-INVIS for the Origin Coset: All Configurations.) [Conditional on O-CRYST.LAT discharged (lattice translation lawful in the Book III ledger).] *For any admissible origin shift $\mathbf{t}$, the translated configuration $X + \mathbf{t}$ satisfies $F_{X+\mathbf{t}}(hkl) = e^{-2\pi i \mathbf{h} \cdot \mathbf{t}} F_X(hkl)$: every intensity is unchanged while every phase shifts by the linear form $\phi(hkl) \mapsto \phi(hkl) - 2\pi \mathbf{h} \cdot \mathbf{t}$. The phase profile is therefore quotient-invisible along the origin coset: intensity-only probes determine phases at best modulo the linear-phase action.*

PROOF. Direct computation on the certified structure factors: a common translation of all scatterers multiplies each $F_X(hkl)$ by the unit-modulus factor $e^{-2\pi i \mathbf{h} \cdot \mathbf{t}}$, which cancels in $|F|^2$. Lawfulness of the translation as an admissible re-description is O-CRYST.LAT. Since the two configurations are admissible re-descriptions of one another with identical intensity profiles and distinct phase profiles (for $\mathbf{t}$ not a lattice vector), the linear-phase coset is quotient-invisible. Combined with Thm. 12.10, the certified invisible group for Friedel-symmetric intensity-only probing of non-centrosymmetric configurations contains the origin translations together with conjugation: phases are invisible at least up to $\phi(\mathbf{h}) \mapsto \pm\phi(\mathbf{h}) - 2\pi \mathbf{h} \cdot \mathbf{t}$. $\square$ $\square$

THEOREM 12.12 ([C]). (PHASE-RECOV for the Enantiomorph under a Nominated Anomalous Probe Extension.) [Conditional on: nomination of a lawful *anomalous-scattering probe extension* $\mathcal{P}'_{\text{diff}} \supset \mathcal{P}_{\text{diff}}$ — a resonant-wavelength channel declared as a certified observable variable, sourced through $\mathbf{U}$ with no hidden supplement (a calibration-pattern declaration, the same nomination class as the pending diffraction standard); and the template machinery of Thm. 10.5.] *Under $\mathcal{P}'_{\text{diff}}$ the Friedel symmetry of Def. 12.9 is broken: the certified atomic response acquires an imaginary component, the Bijvoet differences $|F_X(hkl)|^2 - |F_X(\overline{hkl})|^2$ are generically nonzero for non-centrosymmetric X , and they change sign under conjugation. The enantiomorph is therefore quotient-visible under $\mathcal{P}'_{\text{diff}}$: PHASE-RECOV holds for the conjugation invariant, and the MSC selects the unique minimal faithful phase-sensitive extension among role-equivalent lawful nominations.*

PROOF. With an imaginary response component the structure factors of X no longer satisfy $F_X(\overline{hkl}) = F_X(hkl)^*$, so the two members of a Friedel pair carry independent certified intensities; their difference is an admissible probe outcome of $\mathcal{P}'_{\text{diff}}$. Under configuration inversion the Bijvoet difference changes sign, so X and $\overline{X}$ have distinct $\mathcal{P}'_{\text{diff}}$ profiles whenever at least one certified Bijvoet difference is nonzero — the generic non-centrosymmetric situation, and a finitely checkable condition on the declared reflection set. Hence the conjugation invariant hidden under the Friedel-symmetric discipline (Thm. 12.10) becomes quotient-visible. Lawfulness of the extension is the content of the nominated declaration and is not re-proved here; the selection of the unique minimal faithful extension is the PHASE-RECOV clause of the template (Thm. 10.5) by the Zorn/finite-stabilization argument. The origin coset (Thm. 12.11) remains invisible under this extension, as the linear phase factor cancels in every $|F|^2$ -type channel; origin fixing is a separate declaration, not a probe-information question. $\square$ $\square$

OPEN OBLIGATION 12.13 ([C]). (O-CRYST.PHASE-RECOV: Anomalous-Scattering Wavelength Nominated.) [Conditional on Thm. 12.12 and the declaration below.] A resonant-wavelength anomalous-scattering channel is hereby nominated as the lawful phase-sensitive probe extension $\mathcal{P}'_{\text{diff}}$: an incident-radiation wavelength declared near a certified absorption-edge of a scatterer species present in the configuration, so that the certified atomic response acquires its resonant (imaginary) component. The wavelength enters as a *certified observable variable* — a declared property of the probe process, on the same footing as the SPEC caesium frequency nomination (SPEC Branch, `ob:SPEC-Planck-calibration`) — not as an imported

physical primitive. This extension satisfies the three lawfulness conditions: (a) it is sourced through **U** as a declared probe-response process; (b) it introduces no hidden supplement, the resonant component being a certified response of the already-declared scatterer species, not a new ontology; and (c) its intensity profile (now including the Bijvoet differences) determines the enantiomorph by Thm. 12.12. The numerical wavelength value, when a specific edge is fixed, enters only through the empirical calibration map, not through the foundational derivation — exactly as the caesium value does for $\hbar$.

OPEN OBLIGATION 12.14 ([C]). (O-CRYST.CAL: Diffraction Standard Nominated.) [Conditional on the diffraction calibration pattern and the declaration below.] A characteristic-emission wavelength is hereby nominated as the diffraction length standard for the branch's metric calibration, in the calibration-pattern slot left pending in the closure ledger:

$$\lambda_{\text{cal}} = \lambda(\text{Cu } K\alpha_1) \approx 1.5406 \text{ \AA} \quad (\text{declared empirical reference wavelength}).$$

This plays for CRYST the role the caesium hyperfine frequency plays for SPEC: a declared empirical reference that converts the certified dimensionless diffraction-geometry ledger (Bragg-angle ratios, which are quotient-visible without it) into SI length units. It is sourced through **U** as a certified emission-process observable, adds no hidden supplement, and its numerical value lives strictly in the empirical calibration map: the certified *ratios* of lattice spacings are nomination-independent, and only their SI *scale* is fixed by λ_{cal} . This completes the calibration-pattern row recorded as pending for the diffraction standard.

OPEN OBLIGATION 12.15 ([O]). (O-CRYST.PHASE: Current Status and Remaining Frontier.) Phase problem progress summary after Sessions L+1–L+4:

Conditionally established [C]:

- (1) Certified phase ambiguity: Babinet companion (Thm. 12.2).
- (2) Positivity constrains admissible phases (Thm. 12.5).
- (3) PHASE-INVIS for centrosymmetric configurations up to global sign: the 73 centrosymmetric space groups force real structure factors; global sign not intensity-visible (Thm. 12.8).

Conditionally established this pass [C] (continued):

- (4) PHASE-INVIS for the enantiomorph: conjugation negates all phases while preserving all Friedel-symmetric intensities; absolute configuration quotient-invisible (Thm. 12.10).
- (5) PHASE-INVIS for the origin coset: linear phase factor cancels in every intensity; phases determined at best modulo the origin action (Thm. 12.11).
- (6) PHASE-RECOV for the enantiomorph, conditional on the nominated anomalous-scattering extension: Bijvoet differences make conjugation quotient-visible (Thm. 12.12).

Genuinely open [O] (narrowed):

- (1) *PHASE-INVIS/RECOV modulo the certified invariance group*: for generic non-centrosymmetric configurations, decide whether intensities determine the phases *modulo* the certified invisible group (origin coset together with conjugation), or whether further quotient-invisible phase freedom (homometry) persists generically. This residual is a structure-factor fibration question outside the current NFC collar-algebra toolkit — the same register as **ob:rh-s1-formal** (toolkit-boundary item).
- (2) *Nominations pending*: the concrete anomalous-scattering declaration (resonant wavelength as certified observable variable) and the diffraction standard remain calibration-pattern declarations to be made.

13. Referee Attack Table and Non-Jargon Architecture

REMARK 13.1 ([R]). **Referee attack table (Book VII external falsifiability).** Each step can be refuted by exhibiting the named failure:

Step	Refutation condition
O-CRYST.PR	probe response not sourced through U
O-CRYST.GAP	certified crystalline configuration with $B_{\text{Cryst}}(X) = 0$
O-CRYST.LAT	translation cost not additive in Book III ledger
O-CRYST.NC	fifth intensity-loss channel beyond the declared BLC package
O-CRYST.DOM	$\varepsilon_n^{\text{dom}} \not\rightarrow 0$
O-CRYST.EWALD	$\varepsilon_n^{\text{Ewald}} \not\rightarrow 0$
O-CRYST.SG	MSC selects a carrier outside the 230 space groups
O-CRYST.END	any prior discharge fails
O-CRYST.PHASE	certified intensity-only probe recovers the phase (PHASE-INVIS), or undeclared import in extended probe family (PHASE-RECOV)

REMARK 13.2 ([R]). **Non-jargon proof architecture.** The branch result in standard analytic language:

- (1) *Finite scattering approximants*. Exhaustion $U_1 \subset U_2 \subset \dots$ of $\mathbb{R}^3$ with $|\partial U_n|/|U_n| \rightarrow 0$; certified scattering-density variable; gauge-invariant physical observable $I(hkl) = |F(hkl)|^2 \geq 0$.
- (2) *Certified observables and admissible tests*. Finite probe discipline (intensity responses only, real and nonnegative); translation equivalence from probe data alone; four anti-smuggling conditions (no Euclidean primitive, no Fourier primitive, no atomic species, no chemical formula).
- (3) *Boundary control and no leakage*. Book II unit cell boundary capacity law; Van Hove condition $|\partial U_n|/|U_n| \rightarrow 0$.
- (4) *Bragg gap (core theorem)*. $B_{\text{Cryst}}(X) > 0$ uniformly in n for certified crystalline configurations. Three components: (a) positivity of intensity at peak positions; (b) $I_{\text{peak}} > I_{\text{bg}}$ by non-trivial probe separation; (c) gap persistence from Book III transport (no gap collapse without declared structure-change event).

- (5) *Space group reduction.* Retain only the MSC-selected minimal faithful symmetry carrier; remove non-load-bearing sectors (abelian factors, topological artifacts, hidden chemical classes). MSC carrier coincides with one of the 230 space groups.
- (6) *Domain and Fourier bridges.* $D_{\text{Cryst}} \sim \mathbb{R}^3$ at no-loss scope; Bragg equation and structure factors licensed as effective shorthands; geometric decay $O(e^{-B_{\text{Cryst}} \cdot n})$ for both losses.
- (7) *Phase problem.* Intensity data $\{I(hkl)\}$ is a non-injective function of the structure factors: Babinet pairs establish certified phase ambiguity. Positivity constraint from the scattering density restricts the admissible phase space. Full phase determination requires either a genericity argument (full PHASE-INVIS) or an extended probe family (PHASE-RECOV).
- (8) *Endpoint.* Certified Bragg peak structure with positive gap, additive lattice ledger, no-hidden-channel control, Euclidean and Fourier notation licensed, space group classification derived.

The two places where the NFC machinery remains visible: Step 4 (Bragg gap as the uniform positive lower bound, parallel to YM uniform coercive estimate) and Step 6 (domain and Fourier no-loss bridges, parallel to YM continuum no-loss theorem). Everything else is standard analytic and crystallographic language.

14. Updated Closure Ledger and Obligation Register

REMARK 14.1 ([R]). **Post-discharge obligation register.**

PR [C], GAP [C], LAT [C], NC [C], DOM [C], EWALD [C], SG [C], END [C], PHASE [O]. The eight [C] obligations are discharged by Thms. 9.3–9.27. O-CRYST.PHASE remains open.

Genuinely open obligations (3):

- (1) *O-CRYST.PHASE (full PHASE-INVIS or PHASE-RECOV):* Babinet ambiguity [C] and positivity constraint [C] established; full generic phase determination or certified extended-probe recovery not yet proved.
- (2) *Implicitly open:* declaration of a calibration-eligible diffraction standard (the crystallographic analogue of O-SPEC.Planck-calibration). This would certify the declared probe-energy window $\mathcal{W}_{\text{diff}}$ against a specific empirical standard.
- (3) *Structure determination:* even with the phase problem partially resolved, uniquely determining the three-dimensional arrangement of scatterers from the certified intensity and phase data requires a structure-determination theorem that has not yet been promoted to canonical form.

Discharge summary. The crystallography branch has reached conditional CERT-CLOSE on the diffraction-periodicity-symmetry-domain endpoint. Eight of nine original obligations are conditionally discharged. The phase problem is the sole remaining structural frontier.

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